

Complex Analysis – 6

Целые и мероморфные функции: особенности и рост

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Теоретический минимум.

Опр. An isolated singularity of f at a is *removable*, a *pole* or *essential* according to whether $\lim_{z \rightarrow a} f(z)$ exists, is infinite, or does not exist; equivalently according to the shape of the Laurent expansion.

Теорема 1 (Риман, Казорати–Вейерштрасс, Пикар) *A singularity is removable if f is bounded near a ; near an essential singularity the image of every punctured neighbourhood is dense in $\mathbb{C}$ (Casorati–Weierstrass) and in fact omits at most one value (большая теорема Пикара).*

Теорема 2 (оценки Коши и Лиувилля) $|f^{(n)}(a)| \leq \frac{n! M(r)}{r^n}$; hence an entire function with $|f(z)| = O(|z|^m)$ is a polynomial of degree at most m , and a bounded entire function is constant.

Факт 1 Разложения: $\pi \cot \pi z = \frac{1}{z} + \sum_{n \neq 0} \left(\frac{1}{z-n} + \frac{1}{n} \right)$ и $\sin \pi z = \pi z \prod_{n \geq 1} \left(1 - \frac{z^2}{n^2} \right)$ — образцы теорем Миттаг-Леффлера и Вейерштрасса о задании функции её полюсами и нулями.

1. Prove Riemann's removable singularity theorem and the Casorati–Weierstrass theorem. Then solve: (S.T. Yau CSMC 2017) let $f: \mathbb{C} \setminus \{0\} \rightarrow \mathbb{C}$ be holomorphic with $|f(z)| \leq |z|^2 + \frac{1}{|z|^{1/2}}$ for z near 0; determine all such f .
2. (S.T. Yau CSMC 2015) Determine all entire functions f satisfying $|f(z)| \leq |z|^2 |\operatorname{Im} z|^2$ for all sufficiently large $|z|$.
3. (IMS 2008) Let f be an entire function and let ω_1, ω_2 be complex numbers with $\omega_1/\omega_2 \notin \mathbb{Q}$. Prove that if $f(z) = f(z + \omega_1) = f(z + \omega_2)$ for all z , then f is constant.

4. (IMS 2009) Let $A \subset \mathbb{C}$ be closed and countable. Prove that a bounded holomorphic function $f: \mathbb{C} \setminus A \rightarrow \mathbb{C}$ is constant.
5. (S.T. Yau CSMC 2023) Suppose f is holomorphic in a neighbourhood of $\overline{\mathbb{D}}$ except for a pole at a point z_0 with $|z_0| = 1$, and let $\sum_{n \geq 0} a_n z^n$ be its Taylor series at the origin. Show that $\frac{a_n}{a_{n+1}} \rightarrow z_0$.